

ACADEMIC AND PROFESSIONAL APPOINTMENTS

- Since 2025 Postdoctoral researcher at [Universität Duisburg-Essen](#), Germany.
2023 - 2025 Postdoctoral researcher at [Radboud Universiteit](#), The Netherlands.
2020 - 2023 Postdoctoral researcher at [Friedrich-Alexander-Universität Erlangen-Nürnberg](#), Germany.
2019 - 2020 Assistant professor at [Polytechnic Institute of Porto](#), Portugal.
2018 - 2019 Researcher at [Universidade do Minho](#) and [INESC TEC](#), Portugal.
2012 - 2018 Researcher at [Universidade de Aveiro](#), Portugal.

EDUCATION

- 2014 - 2019 Ph.D. in Mathematics, Universidade de Aveiro.
 Topic: “Kleisli dualities and Vietoris coalgebras” advised by [Dirk Hofmann](#).
2010 - 2012 M.Sc. in Mathematics and Applications, Universidade de Aveiro.
2007 - 2010 B.Sc. in Mathematics, Universidade de Aveiro.

PUBLICATIONS

- Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. The only distributive law over the powerset monad is the one you know. *18th International Workshop on Coalgebraic Methods in Computer Science (CMCS)*, 2026.
- Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. Relators and Notions of Simulation Revisited. *Logic in Computer Science (LICS)*, 2025.
- Pedro Nora, Jurriaan Rot, Lutz Schröder, and Paul Wild. Relational Connectors and Heterogeneous Bisimulations. *International Conference on Foundations of Software Science and Computation Structures (FoSSaCS)*, 2025.
- Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. Identity-Preserving Lax Extensions and Where to Find Them. *International Symposium on Theoretical Aspects of Computer Science (STACS)*, 2025.
- Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. A point-free perspective on Lax extensions and Predicate liftings. *Mathematical Structures in Computer Science*, 2024.
- Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. Kantorovich Functors and Characteristic Logics for Behavioural Distances. *International Conference on Foundations of Software Science and Computation Structures (FoSSaCS)*, 2023.
- Jonas Forster, Sergey Goncharov, Dirk Hofmann, Pedro Nora, Lutz Schröder, and Paul Wild. Quantitative Hennessy-Milner Theorems via Notions of Density. *Computer Science Logic (CSL)*, 2023.
- Dirk Hofmann and Pedro Nora. Duality theory for enriched Priestley spaces. *Journal of Pure and Applied Algebra*, 2023.
- Dirk Hofmann and Pedro Nora. Hausdorff Coalgebras. *Applied Categorical Structures*, April 2020.
- Dirk Hofmann, Renato Neves, and Pedro Nora. Limits in categories of Vietoris coalgebras. *Mathematical Structures in Computer Science*, 2019.
- Dirk Hofmann and Pedro Nora. Enriched Stone-type dualities. *Advances in Mathematics*, 2018.
- Dirk Hofmann, Renato Neves, and Pedro Nora. Generating the algebraic theory of $C(X)$: the case of partially ordered compact spaces. *Theory and Applications of Categories*, 2018.

- Dirk Hofmann and Pedro Nora. Dualities for modal algebras from the point of view of triples. *Algebra Universalis*, 2015.
- Dirk Hofmann and Pedro Nora. Some notes on Esakia spaces. *Textos de Matemática (Série B)*, 2014.

PREPRINTS

- Jonas Forster, Lutz Schröder, Paul Wild, Barbara König, and Pedro Nora. Computing Distinguishing Formulae for Threshold-Based Behavioural Distances. 2026.

CONFERENCE TALKS

- The Only Distributive Law Over the Powerset Monad Is the One You Know. *18th International Workshop on Coalgebraic Methods in Computer Science*, 2026.
- Identity-Preserving Lax Extensions and Where to Find Them. *42nd International Symposium on Theoretical Aspects of Computer Science*, 2025.
- Kantorovich Functors and Characteristic Logics for Behavioural Distances. *International Conference on Foundations of Software Science and Computation Structures*, 2023.
- Quantitative Hennessy-Milner Theorems via Notions of Density. *Computer Science Logic*, 2023.
- Quantitative Hennessy-Milner Theorems. *26th International Workshop on Algebraic Development Techniques*, 2022.
- Limits in categories of Vietoris coalgebras. *XI Portuguese Category Seminar*, 2017.
- From Stone to Gelfand with the help of Vietoris. *31st Summer Conference on Topology and its Applications*, 2016.
- Stone-type dualities beyond lattice structures I. *4th Workshop on Categorical Methods in Non-Abelian Algebra*, 2016.
- Some remarks on spectral relations. *Final Workshop on project MONDRIAN*, 2013.
- The Vietoris monad. *3rd Workshop on project MONDRIAN*, 2013.
- Dualities for modal algebras. *21st International Workshop on Algebraic Development Techniques*, 2012.

SEMINARS

- Good Notions of Bisimulation. *University of Birmingham*, 2026.
- Bisimulations for BC-systems. *Universität Duisburg-Essen*, 2025.
- Bisimulations Revisited. *Universität Duisburg-Essen*, 2025.
- Maximally permissive notions of bisimulation. *DutchCATs, Radboud Universiteit*, 2023.
- Are identity-preserving lax extensions unique? *Friedrich-Alexander-Universität Erlangen-Nürnberg*, 2022.
- Hennessy-Milner theorems done wrong! *Friedrich-Alexander-Universität Erlangen-Nürnberg*, 2022.
- The Strange Bird or: Is the pointfree way pointless? *Friedrich-Alexander-Universität Erlangen-Nürnberg*, 2021.
- The Strange Bird or: How I learned to stop worrying and love lollipops. *Universidade de Aveiro*, 2021.
- Hausdorff Coalgebras. *Friedrich-Alexander-Universität Erlangen-Nürnberg*, 2020.
- $H + V$ and everything else. *Universidade de Aveiro*, 2018.

- How I met ultrafilters. Universidade de Aveiro, 2017.
- Duality theory beyond dimension zero. Universidade de Aveiro, 2017.
- Functional representations of monads in duality theory. Universidade de Coimbra, 2015.
- Uniform ordered spaces. Universidade de Aveiro, 2014.
- Kripkes duality and applications. Universidade de Aveiro, 2013.

RESEARCH PROJECTS

- Since 2025 SpeQt — *Spectra of Behavioural Distances and Quantitative Logics* — supervised by [Barbara König](#).
- 2022 - 2023 CoAx — *Coinduction Meets Algebra for the Axiomatization and Algorithmics of System Equivalences* — supervised by [Stefan Milius](#) and [Lutz Schröder](#).
- 2020 - 2022 HighMoon — *A High Level Language for Programming and Specifying Multi-Effect Algorithms* — supervised by [Sergey Goncharov](#) and [Lutz Schröder](#).
- 2018 - 2019 KLEE — *Coalgebraic Modeling and Analysis for Computational Synthetic Biology* — supervised by [Luís Soares Barbosa](#) and [Renato Neves](#).
- 2012 - 2013 MONDRIAN — *Foundations for architectural design: Service certification, dynamic reconfiguration and self-adaptability* — supervised by [Dirk Hofmann](#).

RESEARCH GROUPS

- Since 2025 [Theoretical Computer Science group](#) at [Universität Duisburg-Essen](#).
- Since 2020 CIDMA (Collaborator) — Center for Research & Development in Mathematics and Applications.
- 2023 - 2025 iCIS — Institute for Computing and Information Sciences.
- 2020 - 2023 [Chair of Computer Science](#) at [Friedrich-Alexander-Universität Erlangen-Nürnberg](#).
- 2018 - 2019 INESC TEC — Institute for Systems and Computer Engineering, Technology and Science.
- 2014 - 2020 CIDMA — Center for Research & Development in Mathematics and Applications

RESEARCH VISITS

- Visiting researcher at [York University](#) — at invitation of [Walter Tholen](#), October 2017.

GRANTS

- Project KLEE (FCT) (POCI-01-0145-FEDER-030947) research grant awarded by INESC TEC, 2018.
- Category Theory 2014 conference grant awarded by the University of Cambridge.
- Individual Ph.D. grant awarded by the Portuguese Foundation for Science and Technology (FCT), 2013.
- Ph.D. grant awarded by the MAP consurtium ([MAPi](#)) to enroll in their doctoral program in Computer Science, 2013.
- Project MONDRIAN (FCT) (PTDC/EIA-CCO/108302/200) research grant awarded by CIDMA, 2012.
- 21st WADT conference grant awarded by the University of Salamanca, 2012.

SERVICE

- Member of the organising committee of the main annual conference in [Category Theory in 2015](#).
- Reviewer for the conferences [SBLP 2013](#), [WADT'22](#), [CSL 2026](#), [LICS 2026](#), and [CONCUR 2026](#).

TEACHING

- Seminar on logic and applications. [Universität Duisburg-Essen](#), 2025/2026.
Organising and hosting seminars, and supervising students.
- Seminar on games in computer science. [Universität Duisburg-Essen](#), 2025/2026.
Organising and hosting seminars, and supervising students.
- Category theory and coalgebras. [Radboud Universiteit](#), 2024 and 2025.
Delivering tutorials and designing and assessing coursework and examinations.
- Discrete mathematics. [Polytechnic Institute of Porto](#), 2019/2020.
Delivering lectures, and designing and assessing coursework and examinations.
- Linear algebra. [Polytechnic Institute of Porto](#), 2019/2020.
Delivering lectures, and designing and assessing coursework and examinations.

INTERNSHIP SUPERVISIONS

- Wessel de Weijer, [Radboud Universiteit](#), 2024.

BACHELOR THESIS SUPERVISIONS

- Ibrahim Alsabbagh, [Universität Duisburg-Essen](#), 2026.
- Ujjwal Kuhar, [Universität Duisburg-Essen](#), 2026.

SCHOOLS

- IBM-QuantaLab School on Quantum Computing 2018.
- 22nd European Intensive Course on Complex Analysis, its Generalizations and Applications. Aveiro, 2017.
- Spring School on Logic of Dynamical Systems. Braga, 2014.

LANGUAGES

- Native Portuguese speaker and fluent in English.